

Exam 2: NE533: Nuclear Fuel Performance

Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

1. (10 pts) Consider a fuel rod with a pellet radius of 0.55 cm that is experiencing a linear heat rate of 300 W/cm.
 - a. What is the maximum stress experienced by the pellet? $k_f = 0.025$ W/cm-K, $E = 200$ GPa, $\nu = 0.35$, and $\alpha = 10 \times 10^{-6}$ 1/K?
 - b. Given $\sigma_{fracture} = 175$ MPa, how far do cracks extend into the fuel?
2. (8 pts) Consider the stress state in a zircaloy-clad fuel rod pressurized to 15 MPa with an average radius of 0.55 cm and a cladding thickness of 0.05 cm.
 - a. Calculate all three components of the stress using the thin-walled cylinder approximation.
 - b. Calculate all three components of the stress at the inside wall, assuming a thick-walled cylinder.
3. (16 pts) Calculate the change in the gap thickness due to thermal expansion. Do not perform iterations. $R_f = 0.48$ cm, $t_{gap} = 0.008$ cm, $T_{CO} = 550$ K, $t_{clad} = 0.08$ cm, $k_{fuel} = 0.03$ W/cm-K, $k_{gap} = 0.003$ W/cm-K, $k_{clad} = 0.15$ W/cm-K, $LHR = 250$ W/cm, $\alpha_c = 10 \times 10^{-6}$ 1/K, $\alpha_f = 14 \times 10^{-6}$ 1/K, $T_{ref}(fuel=clad) = 300$ K.
4. (6 pts) What are the hoop and radial stresses at the inner radius of the cladding due to thermal expansion? $\Delta T = 50$ K; $\alpha_c = 15 \times 10^{-6}$; $E = 100$ GPa; $\nu = 0.34$; $t_c = 0.06$ cm; $R_i = 0.55$ cm.
5. (8 pts) Determine the densification. Fission rate = 6×10^{13} fissions/cm³-s, $\Delta \rho_0 = 0.015$, $\beta_D = 5$ MWD/kgU, $\rho(UO_2) = 10.97$ g/cc, $t = 10$ days. Assume $C_D = 1$.
6. (4 pts) Describe both elasticity and plasticity.
7. (8 pts) Define strain hardening. What causes strain hardening?
8. (3 pts) What three things must all fuel performance codes be able to do?
9. (5 pts) How do grains form in the sintering process of fuel?
10. (8 pts) Describe the concept of microstructure-based (mechanistic) fuel performance modeling and why it is beneficial.
11. (8 pts) What is meant by microstructure? Given an example of a processing technique and the impact it has on the microstructure and related properties.
12. (4 pts) Does diffusion occur more quickly inside the grains or on the grain boundaries? Why?
13. (6 pts) What is a driving force for grain growth? What can impede grain growth?
14. (6 pts) Why does fuel densify? Describe this process.

Equations for NE533

$$\begin{aligned}
Q &= E_f \times N_f \times \sigma_f \times \phi \\
\rho c_p \frac{\partial T}{\partial t} &= \nabla \cdot (k \nabla T) + Q \\
LHR &= \pi R^2 Q_{av} \\
T(r) &= \frac{Q(R^2 - r^2)}{4k} + T_s \\
\\
T_0 - T_s &= \frac{Q}{4k} R_f^2 = \frac{LHR}{4\pi k} \\
T_s - T_{IC} &= \frac{Q}{2h_{gap}} R_f = \frac{LHR}{2\pi R_f h_{gap}} \\
T_{IC} - T_{OC} &= \frac{Q}{2h_{clad}} R_f = \frac{LHR}{2\pi R_f h_{clad}} \\
T_{OC} - T_{cool} &= \frac{Q}{2h_{cool}} R_f = \frac{LHR}{2\pi R_f h_{cool}} \\
\\
h_{gap/clad} &= \frac{k_{gap/clad}}{t_{gap/clad}} \\
\\
k_{gap}(He) &= 16 \times 10^{-6} \times T^{0.79} \\
k_{gap}(Xe) &= 0.7 \times 10^{-6} \times T^{0.79} \\
k_{gap}(y) &= (1 - y) \times k_{He} + y \times k_{Xe} \\
\\
LHR \left(\frac{z}{Z_0} \right) &= LHR^0 \cos \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right] \\
T_{cool}(z) - T_{cool}^{in} &= \frac{2\gamma}{\pi} \frac{Z_0 \times LHR^0}{\dot{m} C_p} \times \\
&\quad \left\{ \sin \frac{\pi}{2\gamma} + \sin \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right] \right\} \\
\\
k_{ox} &= \frac{1}{A + BT} \text{ (W/cmK)} \\
\frac{1}{B} \ln \left(\frac{A + BT_0}{A + BT_s} \right) &= \frac{LHR}{4\pi} \\
A &= 3.8 + 200 \times FIMA \text{ (cmK/W)} \\
B &= 0.0217 \text{ (cm/W)} \\
\\
\sigma_\theta &= \frac{pR}{\delta} \\
\sigma_z &= \frac{pR}{2\delta} \\
\sigma_r &= -\frac{p}{2} \\
\\
\sigma_r(r) &= -p \frac{(R_o/r)^2 - 1}{(R_o/R_i)^2 - 1} \\
\sigma_\theta(r) &= p \frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1} \\
\sigma_z(r) &= p \frac{1}{(R_o/R_i)^2 - 1} \\
\\
\sigma &= -E \Delta T \alpha \\
\\
\sigma_r(r) &= \frac{1}{2} \Delta T \frac{\alpha E}{1 - \nu} \left(\frac{r}{R_i} - 1 \right) \left(1 - \frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right) \\
\sigma_\theta(r) &= \frac{1}{2} \Delta T \frac{\alpha E}{1 - \nu} \left(1 - 2 \frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right) \\
\sigma_z(r) &= \frac{1}{2} \Delta T \frac{\alpha E}{1 - \nu} \left(1 - 2 \frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right) \\
\\
\sigma_r(\eta) &= -\sigma^*(1 - \eta^2) \\
\sigma_\theta(\eta) &= -\sigma^*(1 - 3\eta^2) \\
\sigma_z(\eta) &= -2\sigma^*(1 - 2\eta^2) \\
\\
\sigma^* &= \frac{\alpha E (T_0 - T_s)}{4(1 - \nu)} \\
\\
\Delta \delta_{gap} &= \bar{R}_c \alpha_c (\bar{T}_c - T_{ref}) - R_f \alpha_f (\bar{T}_f - T_{ref}) \\
\\
D(t)^2 - D_0^2 &= 2M_{GB} \gamma_{GB} t \\
\\
M_{GB}(T) &= M_0 e^{(-\frac{Q}{k_B T})} \\
\\
\frac{dD}{dt} &= k \left(\frac{1}{D} - \frac{1}{D_m} \right) \\
\\
\epsilon_D &= \Delta \rho_0 \left(e^{\frac{\beta \ln 0.01}{C_D \beta D}} - 1 \right) \\
\\
\beta &= \frac{\dot{F} t}{N_U}
\end{aligned}$$